A note on Open AI's "A counterexample to Hadwiger's conjecture"

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1 Summary

The manuscript constructs a triangle-free graph G whose complement $H = \bar{G}$ satisfies $\alpha(H) \leq 2$ and $\mathrm{cm}(H) < m/100$, where m is the number of vertices. The connected matching number is defined as follows:

Definition 1. *A matching M in a graph H is called a* connected matching *if every two distinct edges of M are joined by an edge of H: for all distinct $e, f \in M$, there exist $u \in V(e)$ and $v \in V(f)$ such that $uv \in E(H)$. The* connected matching number *is*

$$\mathrm{cm}(H) = \max\{|M| : M \text{ is a connected matching in } H\}.$$

The complement graph provides a counterexample to Hadwiger's conjecture, which asserts that $\chi(H) \leq h(H)$, where $h(H)$ is the largest integer t such that H contains a K_t minor. Indeed, Proposition 3.5 of the paper (proved in Appendix A) gives

$$h(H) \leq \frac{|V(H)| + 4\,\mathrm{cm}(H) + 2}{3}.$$

Applying this to $H = \bar{G}$, and using $\alpha(\bar{G}) \leq 2$ so that every color class has at most two vertices, we obtain, for m large enough,

$$h(\bar{G}) \leq \frac{m + 4\,\mathrm{cm}(\bar{G}) + 2}{3} < \frac{26m}{75} + \frac{2}{3} < \frac{m}{2} \leq \chi(\bar{G}).$$

Thus $h(\bar{G}) < \chi(\bar{G})$, contrary to Hadwiger's conjecture.

In summary, we seek a triangle-free graph $G = (V_G, E_G)$ on m vertices with the following property. Define an auxiliary conflict graph $\widetilde{G}$ whose vertices are ordered nonedges of G:

$$\widetilde{V} = \{(a, b) \in V_G^2 : a \neq b,\ \{a, b\} \notin E_G\}.$$

Two distinct vertices $(a_1, a_2), (b_1, b_2) \in \widetilde{V}$ are adjacent in $\widetilde{G}$ precisely when at least one of the following holds:

1. Their pairs share an endpoint: $\{a_1, a_2\} \cap \{b_1, b_2\} \neq \varnothing$.

2. All four cross pairs are edges of G:

$$\{a_i, b_j\} \in E_G \qquad \text{for all } i, j \in \{1, 2\}.$$

Our target is the independence number of $\tilde{G}$ is less than $m/100$.

2 The construction

2.1 Triangle-free graphs from linear algebra

We first describe the general linear-algebraic scheme and then specify the data used in the manuscript. All vector spaces, ranks, and matrix operations are over $\mathbb{F}_2$.

Let V, W be finite-dimensional vector spaces. Fix $a \in V^*$ and a symmetric bilinear form $T : V \times V \to \mathbb{F}_2$, and put

$$r(\lambda) = a + T(\lambda, \cdot) \in V^*.$$

Let Ω be the set of vertices. For each $o \in \Omega$, assign an injective linear map $U_o : V \to W$ and a linear functional $u_o \in W^*$. Then we can define the associated graph:

Definition 2. *Two vertices $i, j \in \Omega$ are adjacent if there exist $\lambda_i, \lambda_j \in V$ such that*

$$U_i\lambda_i = U_j\lambda_j, \tag{1}$$

$$u_j \circ U_i = r(\lambda_i), \qquad u_i \circ U_j = r(\lambda_j), \tag{2}$$

$$a(\lambda_i) + a(\lambda_j) = 1. \tag{3}$$

Equation (2) *is an equality in V^*.*

Lemma 1. *The associated graph is loopless and triangle-free.*

Proof. Loopless is guaranteed by injectivity of U and (3). Suppose i, j, k is a triangle. Write λ_{ij} for the λ_i at i for the pair ij. Evaluate (2) on λ_{ik}:

$$u_j(U_i\lambda_{ik}) = a(\lambda_{ik}) + T(\lambda_{ij}, \lambda_{ik}).$$

Then sum six equations by permuting i, j, k. For fixed j, the two terms on the left cancel because $U_i\lambda_{ik} = U_k\lambda_{ki}$. For fixed i, the two bilinear terms cancel by symmetry of T. The remaining right-hand side is

$$\sum_{\{i,j\}} \big(a(\lambda_{ij}) + a(\lambda_{ji})\big) = 1 + 1 + 1 = 1,$$

whereas the left-hand side is zero, a contradiction.

□

2.2 Construction of U and u

Write $\mathrm{Mat}_{m,n}(\mathbb{F}_2)$ for the space of $m \times n$ binary matrices, and $\mathrm{Mat}_n(\mathbb{F}_2) = \mathrm{Mat}_{n,n}(\mathbb{F}_2)$. Fix a finite index set $\mathcal{E}$, positive integers N, d_B, h with $N \geq 2(d_B + h)$, and a matrix $E \in \mathrm{Mat}_{d_B}(\mathbb{F}_2)$, to be specified later. Take

$$V \subseteq \bigoplus_{e \in \mathcal{E}} \mathrm{Mat}_{d_B}(\mathbb{F}_2) \quad \text{a linear subspace,} \qquad W = \bigoplus_{e \in \mathcal{E}} \mathrm{Mat}_N(\mathbb{F}_2).$$

For $x \in V$, write $x = (x_e)_{e \in \mathcal{E}}$, so $x_e \in \mathrm{Mat}_{d_B}(\mathbb{F}_2)$.

At each component e, consider quadruples

$$P_e, Q_e \in \mathrm{Mat}_{N,d_B}(\mathbb{F}_2), \qquad X_e, Y_e \in \mathrm{Mat}_{N,h}(\mathbb{F}_2)$$

satisfying

$$\begin{aligned} &[P_e\ X_e] \text{ and } [Q_e\ Y_e] \text{ have full column rank,} \\ &P_e^T Q_e = E, \\ &X_e^T Q_e = 0, \qquad P_e^T Y_e = 0. \end{aligned}$$

The condition $N \geq 2(d_B + h)$ ensures there is enough admissible quadruples. Let $\mathcal{O}$ be the set of tuples $o = (P_{o,e}, Q_{o,e}, X_{o,e}, Y_{o,e})_{e \in \mathcal{E}}$ of admissible quadruples. Define a probability law on $\mathcal{O}$ by sampling uniformly at each component, independently across components.

For matrices of the same size, define $\langle A, B \rangle = \mathrm{Tr}(A^T B)$. Each $o \in \mathcal{O}$ determines $U_o : V \to W$ and $u_o \in W^*$ by

$$U_o(x) = (P_{o,e} x_e Q_{o,e}^T)_{e \in \mathcal{E}}, \qquad u_o(\xi) = \sum_{e \in \mathcal{E}} \langle Y_{o,e} X_{o,e}^T, \xi_e \rangle.$$

for $x \in V$ and $\xi \in W$. Each image component $P_{o,e} x_e Q_{o,e}^T$ belongs to $\mathrm{Mat}_N(\mathbb{F}_2)$. Left inverses of $P_{o,e}$ and $Q_{o,e}$ recover x_e from that component, so U_o is injective. Thus these maps provide the vertex data required in Section 2.1; the particular subspace V and the forms a, T will be specified below.

Moreover,

$$u_o(U_o(x)) = \sum_{e \in \mathcal{E}} \mathrm{Tr}(X_{o,e} Y_{o,e}^T P_{o,e} x_e Q_{o,e}^T) = 0,$$

since $Y_{o,e}^T P_{o,e} = 0$.

2.2.1 A model for uniformly generating quadruples

Set $k = d_B + h$, and combine the columns into

$$A = [P\ X], \qquad B = [Q\ Y] \in \mathrm{Mat}_{N,k}(\mathbb{F}_2).$$

The constraints say that A, B have full column rank and

$$A^T B = H_C, \qquad H_C = \begin{pmatrix} E & 0 \\ 0 & C \end{pmatrix}, \qquad C \in \mathrm{Mat}_h(\mathbb{F}_2).$$

Here $C = X^T Y$ is unrestricted. The letter B in this subsection denotes the combined minus frame, not the coefficient space introduced later.

First fix a full-column-rank A. Choose a right inverse D of A^T and a matrix K whose columns form a basis of $\ker A^T$:

$$D \in \mathrm{Mat}_{N,k}(\mathbb{F}_2), \qquad K \in \mathrm{Mat}_{N,N-k}(\mathbb{F}_2), \qquad A^T D = I_k, \quad A^T K = 0.$$

Both can be obtained by Gaussian elimination, using a fixed rule for each A. Every matrix satisfying the linear constraints has a unique expression

$$\boxed{B = DH_C + KZ}, \qquad C \in \mathrm{Mat}_h(\mathbb{F}_2), \quad Z \in \mathrm{Mat}_{N-k,k}(\mathbb{F}_2).$$

Indeed, $A^T B = H_C$ and the remaining columns lie in $\ker A^T$. Since $[D\ K]$ is invertible,

$$\operatorname{rank} B = k \quad \Longleftrightarrow \quad \operatorname{rank} \begin{pmatrix} H_C \\ Z \end{pmatrix} = k \quad \Longleftrightarrow \quad Z|_{\ker H_C} \text{ is injective.}$$

Thus the linear equations have disappeared, leaving only an injectivity condition on the directions not detected by H_C.

To sample uniformly, use the following procedure:

1. Sample A uniformly among full-column-rank matrices in $\mathrm{Mat}_{N,k}(\mathbb{F}_2)$, and compute D, K.

2. Sample C and Z with independent uniform binary entries. If the stacked matrix above does not have rank k, resample both C and Z, keeping A fixed.

3. Set $B = DH_C + KZ$ and split $A = [P\ X]$ and $B = [Q\ Y]$ at column d_B.

For each A, the correspondence between accepted pairs (C, Z) and admissible minus frames B is bijective. Moreover, the acceptance condition depends only on C, Z, E, not on A, so every full-rank A has the same number of admissible completions. Consequently the resulting quadruple is uniform among all admissible quadruples. Repeat this procedure independently over components.

The rejection step has a uniformly bounded expected cost. For fixed C, let $t = \dim \ker H_C$. The restriction of a uniform Z to this kernel is a uniform $(N-k) \times t$ matrix, so its probability of being injective is

$$\prod_{j=0}^{t-1} (1 - 2^{j-(N-k)}),$$

with the empty product equal to one. Since $t \le k \le N-k$, this probability is at least $1/4$: it is bounded below by $\prod_{r=1}^{\infty}(1-2^{-r})$, whose first two factors give $3/8$ and whose remaining product is at least $1-\sum_{r=3}^{\infty}2^{-r}=3/4$. Thus fewer than four trials are needed on average. Importantly, fixing a uniformly sampled C and resampling only Z would generally give a different law, because the acceptance probability depends on $\operatorname{rank} C$.

2.3 Parameters

The only asymptotic parameter is n. The manuscript fixes

$$C_0 = 1000, \qquad g = 10^9 + 1, \qquad D = 4C_0 g, \qquad M = 2^{1000}.$$

The constants b, j_*, J, r_0, M_0 are chosen in the dependency order of Appendix A of the manuscript, before n tends to infinity. Their numerical values are not specified. Set

$$h = 1000 r_0, \qquad N = M_0 n, \qquad m = 2^{C_0 g N}.$$

Here M is a density bound used in the distribution theorem, whereas M_0 scales the ambient dimension. Choose M_0 sufficiently large to satisfy the dimension and counting requirements, including $N \ge 2(d_B + h)$ for all sufficiently large n. We take $n \ge 2r_0$ and $j_* \ge 1$.

2.4 Boolean moment spaces

Let

$$I = \mathbb{Z}/g\mathbb{Z}, \quad \mathcal{E} = \binom{I}{2}, \quad I_d = \{d+1, \dots, d+(g-1)/2\} \subset I.$$

Elements of I are *tags* and elements of $\mathcal{E}$ are *components*. A *Boolean point* consists of a *selector* $s \in \mathbb{F}_2^b$ and *base coordinates* z in g^2+3 blocks of n bits:

$$z = \left((z_{O_{d,t}})_{d,t\in I},\ z_S,\ z_{\#},\ z_Z\right) \in (\mathbb{F}_2^n)^{g^2+3}.$$

For each block label $A \in \{O_{d,t} : d, t \in I\} \cup \{S, \#, Z\}$, write

$$z_A = (z_{A,1}, \dots, z_{A,n}) \in \mathbb{F}_2^n.$$

Thus z has g^2+3 blocks, each containing n bits; $z_{A,k}$ denotes the kth bit in block A. Write

$$p_* = \sum_{j=0}^{j_*} \binom{b}{j}, \qquad B = \mathbb{F}_2^{p_*} \otimes \mathbb{F}_2^{1+(g^2+3)n}, \qquad d_B = \dim B.$$

Let $p(s) \in \mathbb{F}_2^{p_*}$ be the bit vector listing the evaluations at s of all squarefree selector monomials of degree at most j_*, in a fixed order. Define the evaluation vector and its point moment by

$$v(s,z) = p(s) \otimes (1, z), \qquad w(s,z) = v(s,z)v(s,z)^T \in \operatorname{Mat}_{d_B}(\mathbb{F}_2).$$

Identify $B \otimes B$ with $\mathrm{Mat}_{d_B}(\mathbb{F}_2)$ using the fixed coefficient basis. Let $\mathcal{M} \subseteq \mathrm{Mat}_{d_B}(\mathbb{F}_2)$ be the span of all point moments. For $\ell \in I$, let $\mathcal{M}_\ell$ be the span of those point moments whose base values satisfy $O_{d,t} = 0$ whenever $\ell \notin I_d$. Let $\mathcal{M}_*$ be the span obtained by setting all O-blocks to zero.

Define the linear cut-profile map

$$\begin{aligned} f : \bigoplus_{\ell \in I} \mathcal{M}_\ell &\longrightarrow \bigoplus_{e \in \mathcal{E}} \mathcal{M}, \\ (w_\ell)_{\ell \in I} &\longmapsto (w_\ell + w_{\ell'})_{\{\ell,\ell'\} \in \mathcal{E}}. \end{aligned}$$

Take the witness space in Sections 2.1 and 2.2 to be $V = \operatorname{im} f$. A tuple $(w_\ell)_{\ell \in I}$ is a representation of $x \in V$ if $f((w_\ell)_{\ell \in I}) = x$. By Lemma 2.3 of the manuscript,

$$\ker f = \{(w_*)_{\ell \in I} : w_* \in \mathcal{M}_*\},$$

that is, two representations of the same x differ by a constant tuple with value in $\mathcal{M}_*$.

2.5 The functional a, the bilinear form T, and the matrix E

The witness space is now fixed as $V = \operatorname{im} f$, and the ambient space remains $W = \bigoplus_{e \in \mathcal{E}} \mathrm{Mat}_N(\mathbb{F}_2)$ from Section 2.2. We next specify $a \in V^*$, a symmetric bilinear form T on V, and the matrix E used to constrain the frames.

Define *linear testers* on moments by selecting ordered matrix entries:

$$\begin{aligned} \eta_{d,t}(w(s,z)) &= \sum_{\rho=1}^{r_0} z_{O_{d,t},2\rho-1} z_{O_{d,t},2\rho}, \; d,t \in I \\ \eta_S(w(s,z)) &= \sum_{\rho=1}^{r_0} z_{S,2\rho-1} z_{S,2\rho}. \end{aligned}$$

Put $\eta_O = \sum_{d,t} \eta_{d,t}$ and $\eta = \eta_O + \eta_S$. These are elements of $\mathcal{M}^*$ and take values in $\mathbb{F}_2$.

For a cut profile x represented by (w_ℓ), set

$$a_t(x) = \sum_{\ell,d} \eta_{d,t}(w_\ell), \qquad a(x) = \sum_t a_t(x), \qquad b_t(x) = \sum_{\ell \neq t} \eta_S(w_\ell).$$

These functionals do not depend on the representation: the O-testers vanish on $\mathcal{M}_*$, and a constant change contributes $(g-1)\eta_S(w_*) = 0$ to b_t.

Define

$$T_0(x,y) = a(x)a(y) + \sum_{t \in I} \big(a_t(x) b_t(y) + b_t(x) a_t(y)\big).$$

For each $1 \le j \le J$ and ordered pair $e, f \in \mathcal{E}$, fix endomorphisms L_j^{ef}, R_j^{ef} of B, chosen to satisfy the uniform mixing conditions of Lemma 5.5 of the manuscript. With the entrywise

pairing $\langle A, B\rangle = \mathrm{Tr}(A^T B)$, put

$$B_1(x,y) = \sum_{j=1}^{J} \sum_{e,f\in\mathcal{E}} \left\langle x_e, L_j^{ef} y_f (R_j^{ef})^T \right\rangle,$$
$$T_1(x,y) = B_1(x,y) + B_1(y,x), \qquad T = T_0 + T_1.$$

Then T is symmetric and $T(x,x) = a(x)$. This specifies $r(\lambda) = a + T(\lambda,\cdot)$ in the edge definition.

The matrix $E \in \mathrm{Mat}_{d_B}(\mathbb{F}_2)$ prescribes $P_{o,e}^T Q_{o,e} = E$ in Section 2.2. Let E_η be the sum of the ordered tester matrices, so $\langle E_\eta, w\rangle = \eta(w)$ for $w \in \mathcal{M}$. Choose $M_1, \dots, M_b \in \mathrm{Mat}_n(\mathbb{F}_2)$ along with the mixers as in Lemma 5.5, and define $E_\#$ by

$$v(s,z)^T E_\# v(s',z') = \sum_{\rho=1}^{b} (s_\rho + s'_\rho) z_\#^T M_\rho z'_\#.$$

This is bilinear on B since $j_* \geq 1$. Set $E = E_\eta + E_\#$. Its additional term vanishes on coincident point inputs, so

$$\langle E, w\rangle = \eta(w) \qquad (w \in \mathcal{M}).$$

The Z-block does not occur in E. The mixers and the matrices M_ρ are chosen once for each sufficiently large n, before graph sampling. Their existence with the required rank properties is supplied by Lemma 5.5; arbitrary choices need not give the connected-matching bound.

2.6 The vertex law

With $\mathcal{E}, d_B, N, h, V, a, T, E$ now specified, apply Section 2.2. Let $\mathcal{O}_n$ be the set of admissible frame tuples and let μ_n be its product uniform law across components. An element $o \in \mathcal{O}_n$ determines the maps $U_o : V \to W$ and $u_o \in W^*$ already defined there. In the general scheme of Section 2.1, take $\Omega = \mathcal{O}_n$.

The initial random variable is therefore a tuple of frame matrices. The coefficient space, moment spaces, a, T, and E are fixed before drawing these tuples. In particular, the matrix E alone does not specify the graph: adjacency also depends on V, a, T.

2.7 The sampled graph and the remaining theorem

Sample

$$o_1, \dots, o_m \overset{\text{iid}}{\sim} \mu_n, \qquad m = 2^{C_0 g N}.$$

On the positions $1, \dots, m$, put $ij \in E(G)$ exactly when o_i, o_j satisfy the edge criterion (1)–(3); put $ij \in E(H)$ otherwise. This completes the definition of the random graph. Its edges are determined by the sampled frames, with no further independent edge sampling.

Remark 1. *The initial random variable is the tuple of frame matrices o, not a Boolean point (s, z). The monomial list and moment spaces are fixed globally. Boolean points vary when constructing witnesses; sampling point parameters with the frames fixed is called a query in the manuscript.*

Triangle-freeness of G, and hence $\alpha(H) \le 2$, holds for every sample. The harder assertion is that $\mathrm{cm}(H) < m/100$ with probability $1 - \exp(-\Omega(m))$ (Proposition 3.4). To obtain it, the manuscript calls an ordered pair of raw vertices that do not satisfy the edge criterion a *unit*; such a pair represents an edge of H when placed at distinct sampled positions. Two units conflict if all four cross pairs are edges of G, so no edge of H joins their endpoint sets. Their corresponding edges therefore cannot coexist in a connected matching.

Theorem 3.1 asserts that every law σ on units satisfying the pointwise density bounds

$$\sigma_1 \le M\mu_n, \qquad \sigma_2 \le M\mu_n, \qquad \sigma \le 2^{DN}(\mu_n \otimes \mu_n)$$

gives a conflict between two independent σ-units with probability at least 2^{-100gN}. The endpoints within one unit may be dependent. An entropy-based container argument transfers this statement to the sampled graph. Thus the construction above specifies the graph; the subsequent distribution theorem and container argument establish the small connected matching number needed for the counterexample.

3 An abstract container lemma

The container argument does not depend on the linear-algebraic construction. Its inputs are a finite probability space, a family of ordered pairs, and a graph describing which pairs conflict.

Lemma 2 (Containers for pairs with bounded densities)**.** *Let Ω be a finite set with a strictly positive probability measure μ, let $\mathcal{U} \subseteq \Omega^2$, and let Γ be a finite simple graph on $\mathcal{U}$. Fix $\alpha, \beta \ge 1$ and $0 < \varepsilon < 1$. Write $\mu^2 = \mu \otimes \mu$, and call a probability law ρ on $\mathcal{U}$ feasible if*

$$\rho_1 \le \alpha\mu, \qquad \rho_2 \le \alpha\mu, \qquad \rho \le \beta\mu^2$$

pointwise, where ρ_1, ρ_2 are its endpoint marginals. Suppose every feasible law satisfies

$$\Pr_{v,w \,\mathrm{iid}\sim\rho}(vw \in E(\Gamma)) \ge \varepsilon.$$

Set

$$L = 1 + \left\lceil \frac{\log \beta}{-\log(1-\varepsilon)} \right\rceil.$$

Then there is a family $\mathcal{C}$ of at most $(|\mathcal{U}| + 1)^L$ subsets of $\mathcal{U}$ such that:

1. *Every independent set of Γ is contained in some $C \in \mathcal{C}$.*

2. Each $C \in \mathcal{C}$ has associated sets $S_C \subseteq \Omega$ and $F_C \subseteq \Omega^2$ with

$$\mu(S_C) < \alpha^{-1}, \qquad \mu^2(F_C) < \beta^{-1},$$

$$C \subseteq (S_C \times \Omega) \cup (\Omega \times S_C) \cup F_C.$$

Moreover, sample $z_1, \dots, z_m$ independently with law μ. For any integer $1 \le k \le m$, the probability that there exist $2k$ ordered pairs of sampled positions, all using distinct positions, whose types lie in $\mathcal{U}$ and form an independent set in Γ, is at most

$$(|\mathcal{U}| + 1)^L \left[\binom{m}{k} \alpha^{-k} + m^{2k} \beta^{-k} \right]. \tag{4}$$

Repeated pair types are allowed; independence concerns their set of distinct types. Logarithms in this lemma are natural.

Proof. Entropy increase. For a residual family $R \subseteq \mathcal{U}$ supporting a feasible law, let ρ_R minimize

$$\mathcal{D}(\rho \| \mu^2) = \sum_{v \in \mathcal{U}} \rho(v) \log \frac{\rho(v)}{\mu^2(v)}$$

among feasible laws supported on R. The feasible set is compact and convex, so a minimizer exists. Its support contains the support of every feasible law on R: otherwise mixing in such a law gives a negative infinite right derivative at a missing coordinate, contradicting minimality. Differentiating along a segment towards any feasible ν therefore yields the information-projection inequality

$$\mathcal{D}(\nu \| \mu^2) \ge \mathcal{D}(\rho_R \| \mu^2) + \mathcal{D}(\nu \| \rho_R).$$

If ν is supported on $R' \subseteq R$, then

$$\mathcal{D}(\nu \| \rho_R) \ge -\log \rho_R(R').$$

Thus deleting a set of ρ_R-mass at least ε increases the minimum entropy by at least $-\log(1-\varepsilon)$, provided feasibility remains. Every feasible law has entropy between zero and $\log \beta$.

The container procedure. Fix an independent set I. Start with $R = \mathcal{U}$. While feasibility remains, choose a vertex $v \in R$ maximizing $\rho_R(N_R(v))$, making choices deterministic. The hypothesis implies

$$\sum_{v \in R} \rho_R(v) \rho_R(N_R(v)) \ge \varepsilon,$$

so the selected neighborhood has mass at least ε. If $v \in I$, record v and delete $N_R(v)$; otherwise delete v. Each operation preserves $I \subseteq R$ and removes at least one vertex. Stop when R supports no feasible law.

Each recording that leaves feasibility increases the minimum entropy by at least $-\log(1-\varepsilon)$; all other deletions cannot decrease it. Allowing one final recording, there are at most L recorded vertices. Their set determines the terminal family by replaying the procedure, recording a selected vertex precisely when it is in this set. A previously recorded vertex has no remaining neighbors, and cannot be selected again while feasibility holds. Hence at most

$$\sum_{j=0}^{\min(L,|\mathcal{U}|)} \binom{|\mathcal{U}|}{j} \le (|\mathcal{U}|+1)^L$$

terminal containers occur.

Covering a terminal family. For a terminal C, put a left and a right copy of Ω in a flow network. Give source-to-left edges capacity $\alpha\mu(a)$, right-to-sink edges capacity $\alpha\mu(b)$, and the edge from left a to right b capacity $\beta\mu(a)\mu(b)$ whenever $(a,b) \in C$. A flow of value one is exactly a feasible law on C. Therefore the maximum flow is less than one. A minimum cut gives sets $L_0, R_0 \subseteq \Omega$ with

$$\alpha\mu(L_0) + \alpha\mu(R_0) + \beta\mu^2\big(C \cap ((\Omega \setminus L_0) \times (\Omega \setminus R_0))\big) < 1.$$

Take $S_C = L_0 \cup R_0$ and $F_C = C \cap ((\Omega \setminus L_0) \times (\Omega \setminus R_0))$. These satisfy the claimed cover and measure bounds. Fix one such choice for each container.

Sampling. For a fixed container, the probability that at least k sampled positions lie in S_C is at most $\binom{m}{k}\alpha^{-k}$. A fixed collection of k disjoint ordered position pairs consists of independent μ^2 draws, so the probability that all its types lie in F_C is at most β^{-k}. There are at most m^{2k} such collections.

Outside these two exceptional events, fewer than k sampled positions lie in S_C, and fewer than k disjoint sampled pairs lie in F_C. Any collection of disjoint position pairs whose types lie in C then has at most $2k-2$ members: at most $k-1$ meet S_C, and at most $k-1$ others lie in F_C. In particular it cannot have $2k$ members. Every independent set lies in a container, so a union bound proves (4). □

4 Applying the lemma to Theorem 3.1

Take $\Omega = \mathcal{O}_n$, $\mu = \mu_n$, and let $\mathcal{U}$ be the ordered pairs that do not satisfy the edge criterion for G. Define Γ by joining two pairs when all four cross pairs satisfy that criterion. Orienting a connected matching of $H = \bar{G}$ gives disjoint ordered position pairs whose types form an independent set of Γ.

Theorem 3.1 of the manuscript is exactly the hypothesis of Lemma 2 with

$$\alpha = M = 2^{1000}, \qquad \beta = 2^{DN}, \qquad \varepsilon = 2^{-100gN}.$$

Here $\sigma_1(a) = \sum_b \sigma(a,b)$ and $\sigma_2(b) = \sum_a \sigma(a,b)$ are the two endpoint marginals. The theorem applies to every law satisfying the bounds, not just uniform laws. No feasibility assumption on the empirical law of a matching is required.

The fingerprint bound is $L = O(N2^{100gN})$. Since

$$|\Omega| \le 2^{2|\mathcal{E}|N(d_B+h)}, \qquad \log(|\mathcal{U}|+1) = O(N^2),$$

the number of containers is $\exp(O(N^3 2^{100gN})) = \exp(o(m))$, where $m = 2^{C_0 gN}$ and $C_0 = 1000$.

Set $k = \lceil m/200 \rceil$. For each container, the endpoint exception in the sampling proof has probability at most

$$\binom{m}{k} M^{-k} \le (200\mathrm{e}/M)^k = \exp(-\Omega(m)),$$

and the pair exception has probability at most

$$m^{2k} 2^{-DNk} = 2^{(2C_0 g - D)Nk} = 2^{-2C_0 gNk}.$$

Here e is Euler's number and $D = 4C_0 g$. A union bound over $\exp(o(m))$ containers shows that, with probability $1 - \exp(-\Omega(m))$, every collection of disjoint position pairs with independent types has at most $2k - 2 < m/100$ members, by the stronger bound in the sampling proof. Consequently

$$\mathrm{cm}(H) < m/100.$$

The deterministic bound $\alpha(H) \le 2$ gives $\chi(H) \ge m/2$. By Appendix A, for $m \ge 5$,

$$h(H) \le \frac{m + 4\,\mathrm{cm}(H) + 2}{3} < \frac{26m}{75} + \frac{2}{3} < \frac{m}{2} \le \chi(H).$$

Thus Theorem 3.1 implies that the construction produces a counterexample with probability $1 - \exp(-\Omega(m))$ for all sufficiently large n.

A The clique-minor bound (Proposition 3.5)

Proposition 1. *For every finite nonempty graph H with m vertices,*

$$h(H) \le \frac{m + 4\,\mathrm{cm}(H) + 2}{3}.$$

Proof. Write $c = \mathrm{cm}(H)$ and $b = h(H)$. A K_b minor model consists of b disjoint nonempty connected branch sets, with an edge joining every pair. Let s be the number of singleton branch sets and e the number of two-vertex branch sets.

Each two-vertex branch set supplies an edge. The singleton vertices form a clique, so pairing them supplies $\lfloor s/2 \rfloor$ more edges. Together these form a connected matching, since every pair of its edges is joined by an edge of H. Thus

$$c \ge e + \left\lfloor \frac{s}{2} \right\rfloor, \qquad s \le 2(c - e) + 1.$$

Every remaining branch set has at least three vertices, so

$$m \geq s + 2e + 3(b - s - e) = 3b - 2s - e.$$

Consequently,

$$3b \leq m + 2s + e \leq m + 4c - 3e + 2 \leq m + 4c + 2,$$

which proves the claim. □